

# Parametric Study of the Grad-Shafranov Equation

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# Chapter 1

## Introduction

The Grad-Shafranov(GS) equation is a non-linear equilibrium equation used for 2D plasma. The equation can be used to describe the Grad-Shafranov shift. This whereby the plasma is pushed towards the outwards due to an increase in the poloidal beta( $\beta$ ), where  $\beta$  is the ratio particle pressure to magnetic field pressure. The GS equation is given as shown below,

$$\Delta^* \psi = -\mu_0 R^2 \frac{dp}{d\psi} - \frac{1}{2} F \frac{dF}{d\psi} \quad (1.1)$$

Where,

- $\psi$  is the toroidal magnetic flux
- $R$  is the major radius
- $\mu_0$  is the magnetic permeability
- $p$  is the pressure
- $F$  is a function  $F = RB_\phi$  where  $B_\phi$  is the toroidal magnetic field flux (Wilson, 2012)

The poloidal beta can be given as,

$$\beta = \frac{\sum nkT}{\left(\frac{B^2}{2\mu_0}\right)} \quad (1.2)$$

Where  $n$  is the density,  $k$  is the Boltzmann constant and  $T$  is the temperature (Chen, 2016).

An 'o-point' is a point that refers to a point of maximum or minimum magnetic flux. In a fixed boundary, we consider there to be no 'x-points' or points of no intersection between field lines. This also means that the plasma boundary has a fixed shape and that the  $\psi$  is a constant at this boundary.

In this piece of work, the aim is to consider a fixed boundary and solve 1.1. We then investigate the effect of the displacement of the o-point in the plasma( $\Delta R$ ) as a function of the poloidal beta( $\beta$ ).

# Chapter 2

## Method

To solve the Grad-Shafranov equation, a numerical solver is used. The library called FreeGS was used for this purpose. An example given in the documentation of FreeGS was used to solve the GS equation [cite]. To solve the equation, a number of parameters must be provided which include the maximum and minimal radial values, maximum and minimum lengths in the z direction of boundary, the number of grid points in the x and y directions and finally stating that the boundary is fixed and defined. These parameters are the same as the ones given in the documentation. The maximum pressure and plasma current are also provided. The poloidal beta is  $\beta$  is dependent on the maximum pressure. Therefore, the maximum pressure is changed which in turn causes a change in the coordinates of the o-point.

The o-point coordinate looks like  $(R_0, Z)$  and usually the  $R_{geo}$  is given as the geometrical centre of the plasma. The change in the coordinate is calculated by,

$$\Delta R = R_0 - R_{geo} \quad R_{geo} = \frac{R_{min} + R_{max}}{2} \quad (2.1)$$

Using a loop, we iterate between different pressure values and solve the GS equation each time, while recording the o-point in the plasma and the poloidal beta  $\beta$ . The recorded values are then used to plot a graph of  $\Delta R$  as a function of the poloidal beta( $\beta$ ).

The script for this is attached along in this repository.

## Chapter 3

# Results and Discussion

Shown below in figure 3.1 is a plot of  $\Delta R$  as function of the poloidal beta. As observed below, we observe a linear increase in  $\Delta R$  as the poloidal beta is increased. This shows that when  $\beta$  increases the plasma is shifted outwards, that can be interpreted from the increase in the outward migration of the o-point from the geometrical centre. This outward migration of the O-point is the Grad-Shafranov shift. This is also the case if one takes equation 1.1, the term consisting of  $-\mu_0 R^2 \frac{dp}{d\psi}$  increases thereby moving the O-point radially outward.

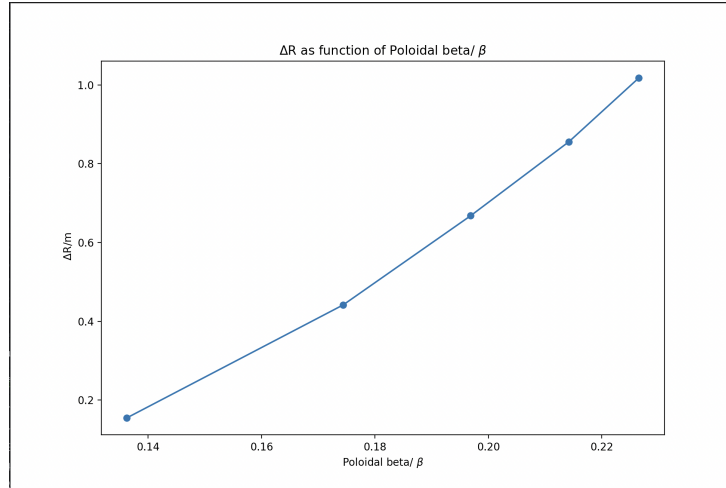


Figure 3.1:  $\Delta R$  given as a function of poloidal beta  $\beta$

A simulation of the magnetic field fluxes are shown below, this have been calculated for 2 pressures of  $1 \times 10^3$  Pa and  $1 \times 10^4$  Pa. The diagrams 3.2 and 3.3 show below the outward movement of the 0-point and the Shafranov shift.

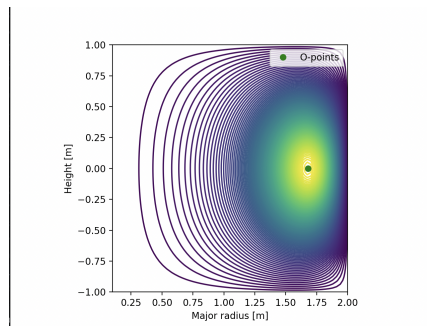


Figure 3.2: Magnetic flux lines with maximum pressure  $1 \times 10^3$  Pa and the poloidal beta of  $\beta = 0.526$

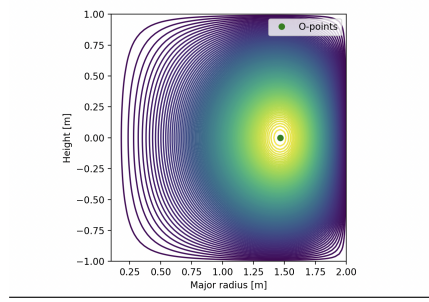


Figure 3.3: Magnetic flux lines with maximum pressure  $1 \times 10^4$  Pa and the poloidal beta of  $\beta = 2.230$

# Bibliography

- Chen, F. F. (2016). Diffusion and resistivity. In *Introduction to plasma physics and controlled fusion* (pp. 145–185). Springer International Publishing. [https://doi.org/10.1007/978-3-319-22309-4\\_5](https://doi.org/10.1007/978-3-319-22309-4_5)
- Wilson, J. I. (2012). Tokamaks, 4th edn., by John Wesson. *Contemporary Physics*, 53(5), 450–451. <https://doi.org/10.1080/00107514.2012.720285>